

Ejercicios Certamen 3 Mecánica Clásica

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1 Problema 1

The transformation of coordinates and momenta has a form

$$P = \alpha \frac{p^2 q}{\sqrt{1 + p^2 q^2}}, \quad Q = \beta \frac{\sqrt{1 + p^2 q^2}}{p}, \quad \alpha, \beta = \text{const} \quad (1.1)$$

- a. Find the values of constants α, β for which the transformation is canonical. For these values of α, β , find explicitly the generating function F which corresponds to the transformation.

Respuesta: Podemos calcular el corchete de Poisson entre $[P, Q]_{p,q}$, es cual deberá ser cero. Usando **Mathematica** tenemos

$$[P, Q]_{p,q} = \alpha\beta \quad (1.2)$$

Por lo tanto, para que la transformación sea canónica deberá cumplirse que

$$\beta = \frac{1}{\alpha} \quad (1.3)$$

Para la ecuación generatriz, buscaremos una ecuación del tipo $F_1(q, Q, t)$ (porque las revise todas y esta se ve la más simple de resolver :p). Escribiendo $p = p(q, Q)$, $P = P(q, Q)$ con **Mathematica**

$$\frac{\partial F_1}{\partial q} = p = \frac{1}{\sqrt{\alpha^2 Q^2 - q^2}}, \quad -\frac{\partial F_1}{\partial Q} = P = \frac{q}{Q\sqrt{\alpha^2 Q^2 - q^2}} \quad (1.4)$$

Usando **Mathematica** para integrar p por q y P por $-Q$ llegamos al mismo resultado

$$F_1(q, Q) = \arctan \left[\frac{q}{\sqrt{\alpha^2 Q^2 - q^2}} \right] \quad (1.5)$$

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- b. Apply the transformation found in the previous item to the hamiltonian of harmonic oscillator $H(p, q) = \frac{p^2}{2m} + \frac{kq^2}{2}$ and find the hamiltonian $\tilde{H}(P, Q)$ in terms of the new variables (P, Q) . Write out the canonical equations in terms of these new variables.

Respuesta: Usando la función generatriz, podemos transformar el hamiltoniano utilizando

$$\tilde{H} = H + \frac{\partial F_1}{\partial t} \quad (1.6)$$

Pero $dF/dt = 0$. Primero, para escribir el hamiltoniano original en función de P, Q , necesitamos $p(P, Q)$, $q(P, Q)$. Usando **Mathematica**

$$q = \frac{\alpha P Q^2}{\sqrt{P^2 Q^2 + 1}}, \quad p = \frac{\sqrt{P^2 Q^2 + 1}}{\alpha Q} \quad (1.7)$$

por lo que el hamiltoniano

$$\tilde{H} = \frac{p^2}{2m} + \frac{kq^2}{2} = \frac{1}{2m\alpha^2} \frac{P^2 Q^2 + 1}{Q^2} + \frac{k\alpha^2}{2} \frac{P^2 Q^4}{P^2 Q^2 + 1} \quad (1.8)$$

Y las ecuaciones canónicas, con **Mathematica**

$$\dot{P} = -\frac{\partial \tilde{H}}{\partial Q} = \frac{1}{\alpha^2 m Q^3} - \frac{\alpha^2 k P^2 Q^3 (P^2 Q^2 + 2)}{(P^2 Q^2 + 1)^2}, \quad \dot{Q} = \frac{\partial \tilde{H}}{\partial P} = \frac{\alpha^2 k P Q^4}{(P^2 Q^2 + 1)^2} + \frac{P}{\alpha^2 m} \quad (1.9)$$

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2 Problema 2

A nonrelativistic charged particle of mass m is constrained to move in a horizontal plane, in the field of the harmonic oscillator $U = kr^2/2$ and the constant magnetic field $\mathbf{B}$ perpendicular to the plane. The vector potential $\mathbf{A}$ may be chosen as

$$\mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r} \quad (2.1)$$

Analyze this system in polar coordinates:

- Write out the lagrangian, the generalized momenta and construct the hamiltonian.

Respuesta: Asumiendo que la partícula tiene carga q , el potencial magnético:

$$U_A = -q\dot{\mathbf{r}} \cdot \mathbf{A} = -q\dot{\mathbf{r}} \cdot \frac{1}{2}(\mathbf{B} \times \mathbf{r})$$

Como $\mathbf{B}$ es perpendicular a $\mathbf{r}$, $\mathbf{B} \times \mathbf{r} = Br\hat{\phi}$

$$U_A = -\frac{1}{2}qBr\dot{\mathbf{r}} \cdot \hat{\phi} = -\frac{qB}{2}r^2\dot{\phi} \quad (2.2)$$

Entonces el lagrangiano

$$L = \frac{m}{2} (\dot{r}^2 + r^2\dot{\phi}^2) - \frac{k}{2}r^2 + \frac{qB}{2}r^2\dot{\phi} \quad (2.3)$$

Y los momentos generalizados

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2\dot{\phi} + \frac{qB}{2}r^2 \quad (2.4)$$

Ya que ϕ no aparece, es cíclica y p_ϕ se conserva. El hamiltoniano entonces

$$\begin{aligned} H &= m\dot{r}^2 + mr^2\dot{\phi}^2 + \frac{qB}{2}r^2\dot{\phi} - \frac{m}{2} (\dot{r}^2 + r^2\dot{\phi}^2) + \frac{k}{2}r^2 - \frac{qB}{2}r^2\dot{\phi} \\ &= \frac{m}{2}\dot{r}^2 + \frac{m}{2}r^2\dot{\phi}^2 + \frac{k}{2}r^2 \end{aligned}$$

Usando los momentos

$$H = \frac{1}{2m} p_r^2 + \frac{1}{2m} \left(\frac{p_\varphi}{r} - \frac{qB}{2} r \right)^2 + \frac{k}{2} r^2 \quad (2.5)$$

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b. Write out the canonical equations of motion.

Respuesta: Las ecuaciones canónicas

$$\dot{p}_r = -\frac{\partial H}{\partial r} = -\frac{1}{m} \left(\frac{p_\varphi}{r} - \frac{qB}{2} r \right) \left(-\frac{p_\varphi}{r^2} - \frac{qB}{2} \right) - kr = \frac{1}{m} \frac{p_\varphi^2}{r^3} - \left[\frac{q^2 B^2}{4m} + k \right] r \quad (2.6)$$

$$\dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 0 \quad (2.7)$$

$$\dot{r} = \frac{\partial H}{\partial p_r} = \frac{1}{m} p_r \quad (2.8)$$

$$\dot{\varphi} = \frac{\partial H}{\partial p_\varphi} = \frac{1}{m} \left(\frac{p_\varphi}{r} - \frac{qB}{2} r \right) \cdot \frac{1}{r} = \frac{1}{m} \frac{p_\varphi}{r^2} - \frac{qB}{2m} \quad (2.9)$$

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c. Write out the equation of Hamilton-Jacobi and integrate it using separation of variables.

Respuesta: La ecuación de Hamilton-Jacobi

$$\frac{\partial S}{\partial t} + \frac{1}{2m} \left(\frac{\partial S}{\partial r} \right)^2 + \frac{1}{2m} \left(\frac{1}{r} \frac{\partial S}{\partial \varphi} - \frac{qB}{2} r \right)^2 + \frac{k}{2} r^2 = 0 \quad (2.10)$$

Como el hamiltoniano no depende del tiempo, el sistema es conservativo y la acción se puede escribir como

$$S = f_r(r) + f_\varphi(\varphi) - Et \quad (2.11)$$

donde

$$\frac{\partial f_\varphi}{\partial \varphi} = p_\varphi = \text{const} \implies f_\varphi = \varphi p_\varphi \quad (2.12)$$

Por lo que

$$\begin{aligned} \frac{1}{2m} \left(\frac{\partial f_r}{\partial r} \right)^2 + \frac{1}{2m} \left(\frac{1}{r} \frac{\partial f_\varphi}{\partial \varphi} - \frac{qB}{2} r \right)^2 + \frac{k}{2} r^2 &= E \\ \frac{1}{2m} \left(\frac{\partial f_r}{\partial r} \right)^2 + \frac{1}{2m} \left(\frac{1}{r} p_\varphi - \frac{qB}{2} r \right)^2 + \frac{k}{2} r^2 &= E \\ \implies \left(\frac{\partial f_r}{\partial r} \right)^2 &= 2mE + qBp_\varphi - \left[\frac{q^2 B^2}{4} + mk \right] r^2 - \frac{1}{r^2} p_\varphi^2 \end{aligned}$$

Resolviendo la integral

$$f_r(r) = \int^r \sqrt{2mE + qBp_\varphi - \left[\frac{q^2 B^2}{4} + mk \right] r^2 - \frac{1}{r^2} p_\varphi^2} dr$$

Por lo que

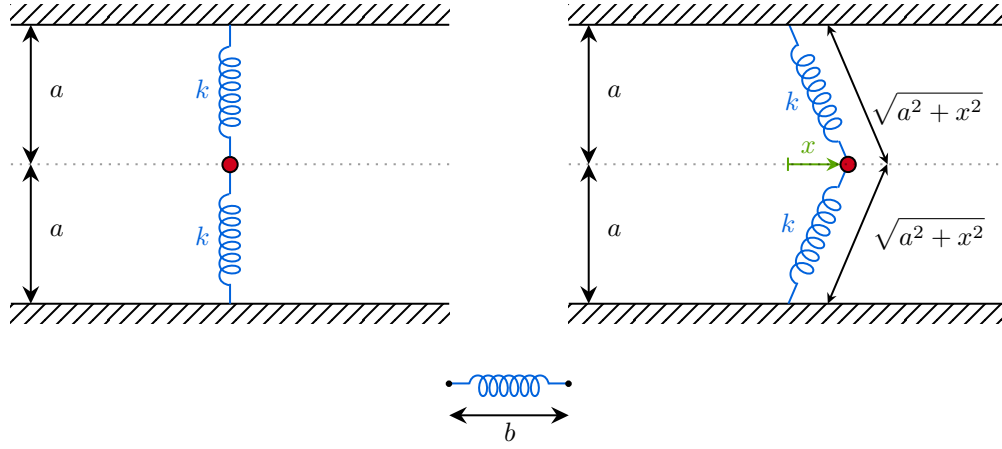
$$S = f_r(r) + \varphi p_\varphi - Et \quad (2.13)$$

$$\Rightarrow \frac{\partial S}{\partial E} = -t_0 = \text{const}, \quad \Rightarrow \frac{\partial S}{\partial p_\varphi} = -\varphi_0 = \text{const} \quad (2.14)$$

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3 Problema 3

A particle of mass m is constrained to move along a horizontal straight rail without friction. Two springs of elasticity k are attached to the spring as shown in the Figure. The lengths of unstretched springs $b < a$. The system might oscillate near its position of equilibrium.



Assuming that $\Delta H = H - H_0$ is a small perturbation, and using canonical perturbation theory, calculate:

- The first nonvanishing correction to the trajectory of the particle

Respuesta: El potencial de este problema

$$\begin{aligned} U(x) &= \frac{1}{2}k \left(\sqrt{a^2 + x^2} - b \right)^2 + \frac{1}{2}k \left(\sqrt{a^2 + x^2} - b \right)^2 \\ &= k \left(\sqrt{a^2 + x^2} - b \right)^2 \end{aligned} \quad (3.1)$$

Por lo que el hamiltoniano

$$H = \frac{1}{2m}p^2 + k \left(\sqrt{a^2 + x^2} - b \right)^2 \quad (3.2)$$

Expandiendo el binomio cuadrado a los 2 primeros órdenes no constantes

$$H \approx \underbrace{\frac{1}{2m}p^2 + k \frac{a-b}{a}x^2}_{H_0} + \underbrace{k \frac{b}{4a^3}x^4}_{\Delta H} \quad (3.3)$$

Para H_0 , las ecuaciones canónicas

$$\begin{aligned}\dot{p}_0 &= -\frac{\partial H}{\partial x} = -2k \frac{a-b}{a} x, & \dot{x}_0 &= \frac{\partial H}{\partial p} = \frac{1}{m} p \\ \implies \ddot{x}_0 &= -\frac{2k}{m} \frac{a-b}{a} x_0\end{aligned}$$

por lo que la solución es un oscilador armónico

$$x_0(t) = A \sin[\omega_0 t + B], \quad p_0(t) = mA\omega_0 \cos[\omega_0 t + B], \quad \omega_0 = \sqrt{\frac{2k}{m} \frac{a-b}{a}} \quad (3.4)$$

Donde A, B vienen de la integración. Utilizando **Mathematica** tenemos

$$A = \frac{\sqrt{m^2 x^2 \omega_0^2 + p^2}}{m\omega_0}, \quad B = \arctan\left(\frac{p}{mx\omega_0}\right) - \omega_0 t \quad (3.5)$$

Para hacer que A, B sean un par de variables α, β canónicas estas deben cumplir con $[\alpha, \beta] = 1$. En nuestro caso

$$[A, B] = \frac{1}{\sqrt{m^2 x^2 \omega_0^2 + p^2}} = \frac{1}{m\omega_0 A} \quad (3.6)$$

Para encontrar el par canónico α de B tendremos que resolver

$$[\alpha(A), B] = \frac{d\alpha}{dA} \cdot [A, B] = 1 \implies \frac{d\alpha}{dA} = m\omega_0 A \quad (3.7)$$

$$\implies \alpha = \frac{m\omega_0 A^2}{2} \quad (3.8)$$

Por lo que obtenemos un par canónico, tal que

$$\alpha = \frac{m\omega_0 A^2}{2}, \quad \beta = B \quad (3.9)$$

Con α relacionado a la amplitud y β a la fase y frecuencia. Reemplazando todo en el hamiltoniano de perturbación

$$\begin{aligned}\Delta H &= k \frac{b}{4a^3} (A \sin[\omega_0 t + B])^4 = k \frac{b}{4a^3} \left(\sqrt{\frac{2\alpha}{m\omega_0}} \sin[\omega_0 t + \beta] \right)^4 \\ \implies \Delta H &= \frac{bk}{a^3 m^2 \omega_0^2} \alpha^2 \sin^4[\omega_0 t + \beta]\end{aligned} \quad (3.10)$$

Y las ecuaciones canónicas para α, β usando **Mathematica**

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{4bk}{a^3 m^2 \omega_0^2} \alpha^2 \sin^3(\beta + \omega_0 t) \cos(\beta + \omega_0 t) \quad (3.11)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{2bk}{a^3 m^2 \omega_0^2} \alpha \sin^4(\beta + \omega_0 t) \quad (3.12)$$

Tomando $\alpha \approx \alpha_0 + \alpha_1(t)$, $\beta \approx \beta_0 + \beta_1(t)$ tenemos

$$\dot{\alpha}_1 \approx -\frac{4bk}{a^3 m^2 \omega_0^2} \alpha_0^2 \sin^3(\beta_0 + \omega_0 t) \cos(\beta_0 + \omega_0 t) \quad (3.13)$$

$$\dot{\beta}_1 \approx \frac{2bk}{a^3 m^2 \omega_0^2} \alpha_0 \sin^4(\beta_0 + \omega_0 t) \quad (3.14)$$

Integrando

$$\alpha_1 \approx -\frac{bk}{a^3 m^2 \omega_0^3} \alpha_0^2 \sin^4(\beta_0 + \omega_0 t) \quad (3.15)$$

$$\beta_1 \approx \frac{3bk}{4a^3 m^2 \omega_0^3} \alpha_0 [\beta_0 + \omega_0 t] + \frac{bk}{16a^3 m^2 \omega_0^3} \alpha_0 [\sin(4\beta_0 + 4\omega_0 t) - 8\sin(2\beta_0 + 2\omega_0 t)] \quad (3.16)$$

Reemplazando en la ecuación de trayectoria,

$$\begin{aligned} x_0(t) &= A \sin[\omega_0 t + B] \\ &= \sqrt{\frac{2\alpha}{m\omega_0}} \sin[\omega_0 t + \beta] \\ &= \sqrt{\frac{2\alpha_0 + 2\alpha_1}{m\omega_0}} \sin[\omega_0 t + \beta_0 + \beta_1] \\ &\approx \left(\sqrt{\frac{2\alpha_0}{m\omega_0}} + \sqrt{\frac{1}{2m\omega_0\alpha_0}} \alpha_1 \right) (\sin[\omega_0 t + \beta_0] + \beta_1 \cos[\omega_0 t + \beta_0]) \\ &\approx \left(\sqrt{\frac{2\alpha_0}{m\omega_0}} - \sqrt{\frac{m\omega_0\alpha_0^3}{2}} \frac{bk \sin^4[\omega_0 t + \beta_0]}{a^3 m^3 \omega_0^4} \right) \left(\sin[\omega_0 t + \beta_0] + \underbrace{\beta_1 \cos[\omega_0 t + \beta_0]}_{\Gamma} \right) \\ &\approx \sqrt{\frac{2\alpha_0}{m\omega_0}} \sin[\omega_0 t + \beta_0] - \sqrt{\frac{\alpha_0^3}{2m^5\omega_0^7}} \frac{bk}{a^3} \sin^5[\omega_0 t + \beta_0] + \sqrt{\frac{2\alpha_0}{m\omega_0}} \Gamma \\ \Gamma &= \frac{\alpha_0 bk \{12[\omega_0 t + \beta_0] - 8\sin(2[\omega_0 t + \beta_0]) + \sin(4[\omega_0 t + \beta_0])\}}{16a^3 m^2 \omega_0^3} \cos[\omega_0 t + \beta_0] \end{aligned} \quad (3.17)$$

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b. The first nonvanishing correction to the frequency of oscillations.

Respuesta:

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